
NEUROLINK WEAR

Predictive health intelligence,
worn every day

SMART WEARABLE + AI/ML + MOBILE SAFETY

Concept-stage pitch deck



Critical health changes often happen between check-ins

02

A wearable can close the gap between an isolated reading and a timely response.

1

Signals lack context

Heart rate, oxygen, temperature and stress indicators become more useful when patterns are tracked together over time.

2

Early warning is difficult

Fatigue, fever, panic and low oxygen may begin as subtle deviations rather than a single obvious event.

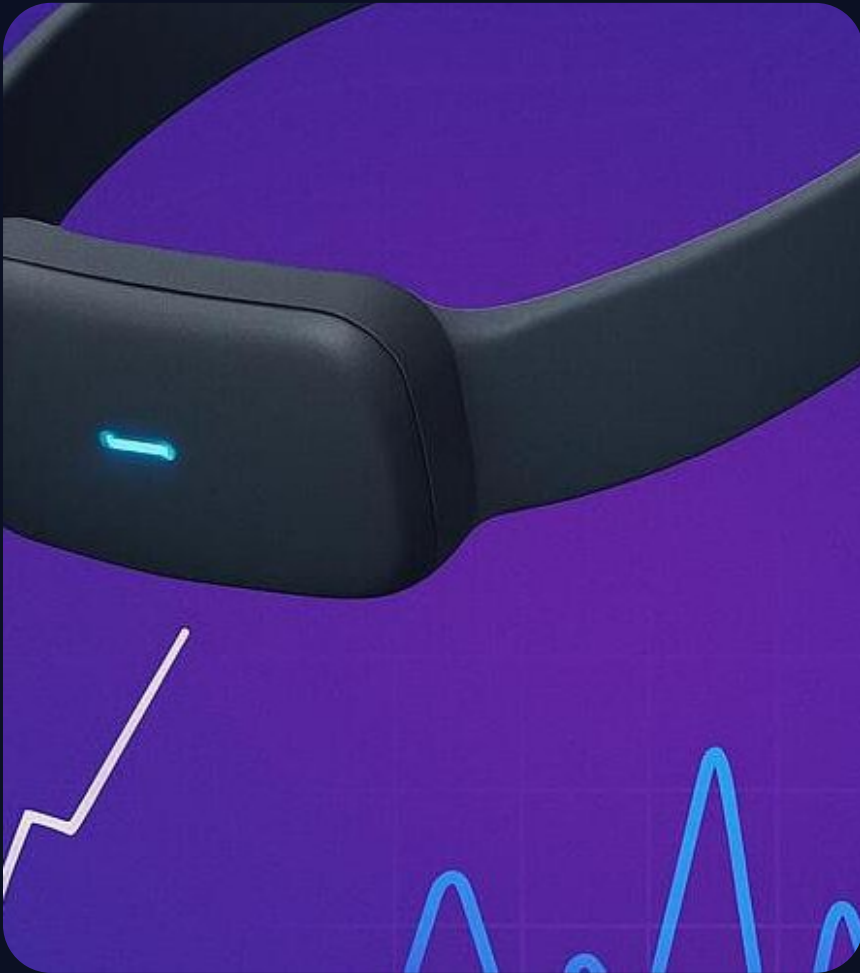
3

Emergencies need escalation

A fall or dangerous reading can require an alert to the wearer and an automatic call to a trusted contact.

One system senses, explains and responds

03



1

SENSE

PPG, GSR, motion, temperature and GPS create a continuous view of the wearer.

2

UNDERSTAND

Edge features and anomaly models identify changes, trends and risk patterns.

3

RESPOND

The app explains the signal, recommends an action and escalates danger.

System architecture

04

A continuous pipeline converts raw sensor data into understandable guidance and safety actions.



MQTT / Wi-Fi every 1-5 seconds

EDGE

RMSSD, variance and trend features

FALLBACK

Statistical threshold rules

OPTIONAL

Autoencoder or LSTM time-series models

Five sensors create a richer view of risk

Each signal answers a different question. The AI layer combines them when a single metric is not enough.

SENSOR	MEASUREMENT	ROLE IN THE SYSTEM
MAX30102 (PPG)	Heart rate, SpO ₂ , HRV	Cardiovascular context, oxygen and recovery signals
GSR	Skin conductance	Stress and sympathetic arousal indicator
MPU6050	Motion and acceleration	Movement, tremor and fall detection
MLX90614	Skin temperature	Fever trend and thermal change
NEO-6M	GPS location	Location context for safety escalation

Sensor readings support wellness and safety insights. They do not replace clinical diagnosis.

The intelligence engine combines anomalies, trends and context

06

CONDITION	SIGNAL PATTERN	DETECTION APPROACH
Stress	HRV down · GSR up · temperature up	Combined anomaly score
Panic attack	Heart rate spike · GSR spike	Sudden deviation + no motion
Fever	Temperature rising over time	Continuous trend
Low oxygen	SpO ₂ below threshold	Threshold-based detection
Fatigue	Sustained low HRV	Long-term trend analysis
Fall	Sudden IMU acceleration	Motion anomaly

MODEL STACK

- 1 Isolation Forest**
Primary anomaly model
- 2 Threshold rules**
Fallback logic
- 3 Autoencoder / LSTM**
Optional time-series layer
- 4 LLM**
Explanation and advice

Fall detection can trigger a safety workflow

Designed emergency escalation

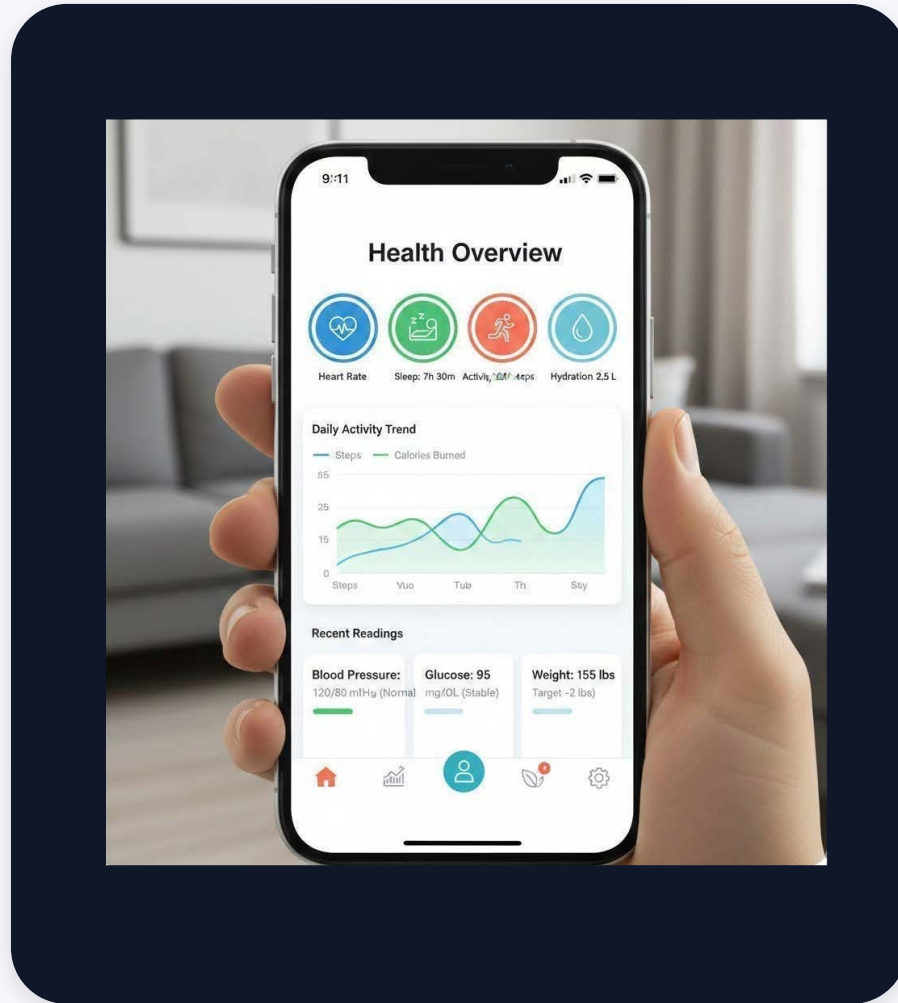
- 1 Detect sudden acceleration or a dangerous vital pattern
- 2 Check inactivity, severity and available location context
- 3 Alert the wearer and request confirmation
- 4 Auto-contact saved emergency numbers when escalation rules are met
- 5 Share the event type, time and GPS location

Escalation thresholds, consent and failure handling require pilot validation.



The mobile app turns live data into clear action

08



USER EXPERIENCE

LIVE Current vital signs and motion status

TRENDS History for HRV, temperature and anomaly patterns

ADVICE LLM-generated explanations and personalized next steps

ALERTS Stress, fever, low oxygen and fall notifications

SAFETY Saved emergency numbers, GPS context and alert history

Initial users and market entry

One sensing platform supports distinct use cases. The first commercial wedge should remain focused.



WELLNESS & STRESS

Young professionals who want context,
not another raw score

INDEPENDENT AGING

Older adults who benefit from fall
detection and proactive monitoring

FAMILY & CAREGIVERS

Trusted contacts who need alerts and
location context

Commercial model and validation path

Proposed hypotheses for a concept-stage startup. Pricing, market size and unit economics remain to be validated.

REVENUE

- 1 DEVICE SALE**
One-time revenue from the wearable and charging hardware
- 2 SUBSCRIPTION**
Recurring access to AI insights, longer history and caregiver safety features
- 3 B2B / B2B2C**
Pilots with senior care, employers, insurers and remote-care partners

VALIDATION ROADMAP

- 01 MVP**
Integrate the sensor stack, edge features, MQTT pipeline and mobile dashboard.
- 02 CONTROLLED PILOT**
Measure detection reliability, false alerts, battery life and user adherence.
- 03 SAFETY & SCALE**
Validate escalation, privacy, regulatory scope, manufacturing and support.

NEXT STEP Secure pilot partners, clinical validation support and early-stage funding to test the complete safety workflow.